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# Deep Residual Learning for Image Recognition
# 图像识别的深度残留学习

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## Abstract 摘要

Deeper neural networks are more difficult to train. We present a residual learning framework to ease the training of networks that are substantially deeper than those used previously. We explicitly reformulate the layers as learning residual functions with reference to the layer inputs, instead of learning unreferenced functions. We provide comprehensive empirical evidence showing that these residual networks are easier to optimize, and can gain accuracy from considerably increased depth. On the ImageNet dataset we evaluate residual nets with a depth of up to 152 layers—8× deeper than VGG nets [41] but still having lower complexity. An ensemble of these residual nets achieves 3.57% error on the ImageNet *test* set. This result won the 1st place on the ILSVRC 2015 classification task. We also present analysis on CIFAR-10 with 100 and 1000 layers.

更深层的神经网络更难训练。我们提出了一个残留学习框架，以简化对比以往使用的网络更深的训练。我们明确将层重新表述为基于层输入的学习残差函数，而不是学习无引用函数。我们提供了全面的实证证据，表明这些残差网络更容易优化，并且可以通过大幅增加的深度获得准确性。 在 ImageNet 数据集中，我们评估深度最多为 152 层的残余网——比 VGG 网 × 更深 8 层 [41]，但复杂度仍然较低。这些残差网的一组在 ImageNet *测试*集上误差为 3.57%。这一结果使他获得了 2015 年 ILSVRC 分类任务的第一名。我们还对 CIFAR-10 进行了 100 层和 1000 层的分析。

The depth of representations is of central importance for many visual recognition tasks. Solely due to our extremely deep representations, we obtain a 28% relative improvement on the COCO object detection dataset. Deep residual nets are foundations of our submissions to ILSVRC & COCO 2015 competitions[1]
表征深度对许多视觉识别任务至关重要。仅凭借极其深度的表征，我们在 COCO 对象检测数据集上获得了 28%的相对提升。深残余网是我们提交 ILSVRC 和 COCO 2015 竞赛作品的基础。, where we also won the 1st places on the tasks of ImageNet detection, ImageNet localization, COCO detection, and COCO segmentation.
我们还在 ImageNet 检测、ImageNet 定位、COCO 检测和 COCO 分割等任务中获得第一名。

[1] http://image-net.org/challenges/LSVRC/2015/ and http://mscoco.org/dataset/#detections-challenge2015.
http://image-net.org/challenges/LSVRC/2015/ 和 http://mscoco.org/dataset/#detections-challenge2015。

## 1 Introduction 1 简介

Deep convolutional neural networks [22, 21] have led to a series of breakthroughs for image classification [21, 50, 40]. Deep networks naturally integrate low/mid/high-level features [50] and classifiers in an end-to-end multi-layer fashion, and the “levels” of features can be enriched by the number of stacked layers (depth). Recent evidence [41, 44] reveals that network depth is of crucial importance, and the leading results [41, 44, 13, 16] on the challenging ImageNet dataset [36] all exploit “very deep” [41] models, with a depth of sixteen [41] to thirty [16]. Many other nontrivial visual recognition tasks [8, 12, 7, 32, 27] have also greatly benefited from very deep models.

深度卷积神经网络[22， 21] 带来了一系列图像分类的突破 [21， 50， 40]。深度网络自然地以端到端多层方式整合低/中/高层特征 [50] 和分类器，且特征的“层级”可以通过堆叠层（深度）的数量来丰富。最新证据[41， 44] 显示网络深度至关重要，而在具有挑战性的 ImageNet 数据集 [36] 上的主要结果[41， 44， 13， 16] 均采用了“非常深” [41] 的模型，深度范围为 16 至 30[16]。 许多其他非平凡的视觉识别任务 [8， 12， 7， 32， 27] 也从非常深的模型中获益良多。

Driven by the significance of depth, a question arises: *Is learning better networks as easy as stacking more layers?* An obstacle to answering this question was the notorious problem of vanishing/exploding gradients [1, 9], which hamper convergence from the beginning. This problem, however, has been largely addressed by normalized initialization [23, 9, 37, 13] and intermediate normalization layers [16], which enable networks with tens of layers to start converging for stochastic gradient descent (SGD) with backpropagation [22].

基于深度的重要性，一个问题出现： *学习更好的网络是否像堆叠更多层那么简单?* 回答这一问题的一个障碍是臭名昭著的梯度消失/爆炸问题 [1， 9]，这从一开始就阻碍了收敛。然而，这一问题已被归一化初始化 [23， 9， 37 ， 13] 和中间归一化层 [16] 解决，这些层允许拥有数十层的网络开始收敛，实现随机梯度下降（SGD）并伴随反向传播 [22]。

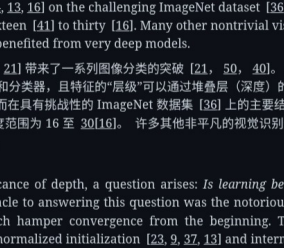


Figure 1: Training error (left) and test error (right) on CIFAR-10 with 20-layer and 56-layer “plain” networks. The deeper network has higher training error, and thus test error. Similar phenomena on ImageNet is presented in Fig. 4.

图 1： CIFAR-10 在 20 层和 56 层“纯”网络上的训练误差（左）和测试误差（右）。更深层的网络训练误差较高，因此测试误差也更高。ImageNet 上的类似现象见图。4.

When deeper networks are able to start converging, a *degradation* problem has been exposed: with the network depth increasing, accuracy gets saturated (which might be unsurprising) and then degrades rapidly. Unexpectedly, such degradation is *not caused by overfitting*, and adding more layers to a suitably deep model leads to *higher training error*, as reported in [11, 42] and thoroughly verified by our experiments. Fig. 1 shows a typical example.

当更深的网络开始收敛时，就暴露出一个*退化*问题：随着网络深度的增加，准确性会被饱和（这并不奇怪），随后迅速下降。出乎意料的是，这种退化*并非由过拟合引起*，向足够深度的模型增加更多层会导致*更高的训练误差*，这一点见 [11， 42] 报道并被我们的实验充分验证。无花果。1 是一个典型例子。

The degradation (of training accuracy) indicates that not all systems are similarly easy to optimize. Let us consider a shallower architecture and its deeper counterpart that adds more layers onto it. There exists a solution *by construction* to the deeper model: the added layers are *identity* mapping, and the other layers are copied from the learned shallower model. The existence of this constructed solution indicates that a deeper model should produce no higher training error than its shallower counterpart. But experiments show that our current solvers on hand are unable to find solutions that are comparably good or better than the constructed solution (or unable to do so in feasible time).

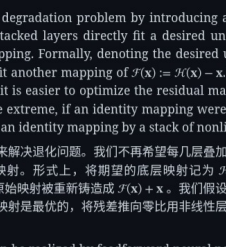
训练准确度的下降表明并非所有系统都同样容易优化。让我们考虑一个较浅的架构及其更深层的对应结构，后者会在上面添加更多层次。存在一个*通过构造*方式解到更深层模型：新增的层是*恒等*映射，其他层则是从学习到的浅模型复制而来。这种构造解的存在表明，深度模型不会产生比浅模型更高的训练误差。但实验显示，我们现有的求解者无法找到与构造解相当或更好的解（或者无法在可行时间内完成）。

In this paper, we address the degradation problem by introducing a *deep residual learning* framework. Instead of hoping each few stacked layers directly fit a desired underlying mapping, we explicitly let these layers fit a residual mapping. Formally, denoting the desired underlying mapping as $\mathcal{H}(\mathbf{x})$, we let the stacked nonlinear layers fit another mapping of $\mathcal{F}(\mathbf{x}):=\mathcal{H}(\mathbf{x})-\mathbf{x}$. The original mapping is recast into $\mathcal{F}(\mathbf{x})+\mathbf{x}$. We hypothesize that it is easier to optimize the residual mapping than to optimize the original, unreferenced mapping. To the extreme, if an identity mapping were optimal, it would be easier to push the residual to zero than to fit an identity mapping by a stack of nonlinear layers.

本文通过引入*深度残差学习*框架来解决退化问题。我们不再希望每几层叠加的图层能直接匹配期望的底层映射，而是明确让这些图层匹配残差映射。形式上，将期望的底层映射记为 $\mathcal{H}(\mathbf{x})$，我们让堆叠的非线性层拟合另一个映射 $\mathcal{F}(\mathbf{x}):=\mathcal{H}(\mathbf{x})-\mathbf{x}$。原始映射被重新铸造成 $\mathcal{F}(\mathbf{x})+\mathbf{x}$。我们假设，优化残差映射比优化原始未被引用映射更容易。极端地说，如果恒等映射是最优的，将残差推向零比用非线性层堆叠拟合恒等映射更容易。

The formulation of $\mathcal{F}(\mathbf{x})+\mathbf{x}$ can be realized by feedforward neural networks with “shortcut connections” (Fig. 2). Shortcut connections [2, 34, 49] are those skipping one or more layers. In our case, the shortcut connections simply perform *identity* mapping, and their outputs are added to the outputs of the stacked layers (Fig. 2). Identity shortcut connections add neither extra parameter nor computational complexity. The entire network can still be trained end-to-end by SGD with backpropagation, and can be easily implemented using common libraries (*e.g.*, Caffe [19]) without modifying the solvers.

的表述 $\mathcal{F}(\mathbf{x})+\mathbf{x}$ 可以通过带有“捷径连接”的前馈神经网络实现（见图）。2）捷径连接 [2， 34， 49] 是指跳过一个或多个层的连接。在我们的案例中，捷径连接仅执行*恒等*映射，其输出被叠加到堆叠层的输出中（见图）。2）恒等捷径连接既不增加额外参数，也不增加计算复杂度。整个网络仍可通过 SGD 通过反向传播端到端训练，且可通过常用库（ *如* Caffe [19]）轻松实现，无需修改求解器。



x
weight layer
relu
weight layer
$\mathcal{F}(\mathbf{x})$
x
identity
$\mathcal{F}(\mathbf{x})+\mathbf{x}$
relu

Figure 2: Residual learning: a building block.

图 2： 残余学习：构建模块。

We present comprehensive experiments on ImageNet [36] to show the degradation problem and evaluate our method. We show that: 1) Our extremely deep residual nets are easy to optimize, but the counterpart “plain” nets (that simply stack layers) exhibit higher training error when the depth increases; 2) Our deep residual nets can easily enjoy accuracy gains from greatly increased depth, producing results substantially better than previous networks.

我们在 ImageNet[36] 上进行了全面的实验，展示退化问题并评估我们的方法。我们证明：1）极深残差网易于优化，但对应的“普通”网（仅堆叠图层）在深度增加时训练误差更高;2）我们的深残差网可以轻松从大幅增加的深度中获得准确率提升，从而产生远超以往网络的结果。

Similar phenomena are also shown on the CIFAR-10 set [20], suggesting that the optimization difficulties and the effects of our method are not just akin to a particular dataset. We present successfully trained models on this dataset with over 100 layers, and explore models with over 1000 layers.

类似现象也出现在 CIFAR-10 集合 [20] 中，表明优化难度和我们方法的影响并非仅与某个特定数据集相同。我们在该数据集上展示了超过 100 个图层的成功训练模型，并探索了超过 1000 个图层的模型。

On the ImageNet classification dataset [36], we obtain excellent results by extremely deep residual nets. Our 152-layer residual net is the deepest network ever presented on ImageNet, while still having lower complexity than VGG nets [41]. Our ensemble has **3.57%** top-5 error on the ImageNet *test* set, and *won the 1st place in the ILSVRC 2015 classification competition*. The extremely deep representations also have excellent generalization performance on other recognition tasks, and lead us to further *win the 1st places on: ImageNet detection, ImageNet localization, COCO detection, and COCO segmentation* in ILSVRC & COCO 2015 competitions. This strong evidence shows that the residual learning principle is generic, and we expect that it is applicable in other vision and non-vision problems.

在 ImageNet 分类数据集 [36] 中，我们通过极深的残差网获得了优异的结果。我们的 152 层残留网是 ImageNet 上有史以来最深的网络，同时复杂度仍低于 VGG 网络 [41]。我们的合奏组在 ImageNet *测试*集中的前五名误差率为 **3.57%**，并在 *2015 年 ILSVRC 分类竞赛中获得第一名*。极其深度的表示在其他识别任务中也表现出色，使我们在 ILSVRC 和 COCO 2015 竞赛中获得 *ImageNet 检测、ImageNet 定位、COCO 检测和 COCO 分割*等项目的第一名。这些强有力的证据表明残差学习原理是通用的，我们预计它适用于其他视觉和非视觉问题。

## 2 Related Work 2 相关工作

**Residual Representations.** In image recognition, VLAD [18] is a representation that encodes by the residual vectors with respect to a dictionary, and Fisher Vector [30] can be formulated as a probabilistic version [18] of VLAD. Both of them are powerful shallow representations for image retrieval and classification [4, 48]. For vector quantization, encoding residual vectors [17] is shown to be more effective than encoding original vectors.

**残余表示。** 在图像识别中，VLAD [18] 是一种通过残差向量相对于字典进行编码的表示，而 Fisher 向量 [30] 可以被表述为 VLAD 的概率版本 [18]。 它们都是图像检索和分类的强大浅表表示 [4， 48]。 对于向量量化，编码残余向量 [17] 比编码原始向量更有效。

In low-level vision and computer graphics, for solving Partial Differential Equations (PDEs), the widely used Multigrid method [3] reformulates the system as subproblems at multiple scales, where each subproblem is responsible for the residual solution between a coarser and a finer scale. An alternative to Multigrid is hierarchical basis preconditioning [45, 46], which relies on variables that represent residual vectors between two scales. It has been shown [3, 45, 46] that these solvers converge much faster than standard solvers that are unaware of the residual nature of the solutions. These methods suggest that a good reformulation or preconditioning can simplify the optimization.

在低层视觉和计算机图形学中，为求解偏微分方程（PDE），广泛使用的多网格方法 [3] 将系统重新表述为多尺度的子问题，每个子问题负责在较粗尺度和较细尺度之间的残差解。多网格的替代方法是分层基础预处理 [45， 46]，它依赖于表示两个尺度间残差向量的变量。研究表明 [3， 45， 46]，这些求解器的收敛速度远快于不了解解的残差性质的标准求解器。这些方法表明，良好的重新表述或预处理可以简化优化。

**Shortcut Connections.** Practices and theories that lead to shortcut connections [2, 34, 49] have been studied for a long time. An early practice of training multi-layer perceptrons (MLPs) is to add a linear layer connected from the network input to the output [34, 49]. In [44, 24], a few intermediate layers are directly connected to auxiliary classifiers for addressing vanishing/exploding gradients. The papers of [39, 38, 31, 47] propose methods for centering layer responses, gradients, and propagated errors, implemented by shortcut connections. In [44], an “inception” layer is composed of a shortcut branch and a few deeper branches.

**快捷连接。** 导致捷径连接的实践和理论[2， 34， 49] 已被长期研究。训练多层感知器（MLPs）的早期做法是从网络输入端添加一个线性层到输出 [34， 49]。 在 [44， 24] 中，有几层中间层直接连接到辅助分类器，用于处理消失/爆炸梯度。 [39， 38， 31， 47] 的论文提出了通过快捷连接实现的层响应、梯度和传播误差中心的方法。 在 [44] 中，“起始”层由一个快捷分支和几个更深层的分支组成。

Concurrent with our work, “highway networks” [42, 43] present shortcut connections with gating functions [15]. These gates are data-dependent and have parameters, in contrast to our identity shortcuts that are parameter-free. When a gated shortcut is “closed” (approaching zero), the layers in highway networks represent *non-residual* functions. On the contrary, our formulation always learns residual functions; our identity shortcuts are never closed, and all information is always passed through, with additional residual functions to be learned. In addition, highway networks have not demonstrated accuracy gains with extremely increased depth (*e.g.*, over 100 layers).

与我们的研究同时，“高速公路网络”[42， 43] 提供了带有门控函数的捷径连接 [15]。 这些门依赖于数据且有参数，而我们的身份捷径则是无参数的。当门控捷径“关闭”（接近零）时，高速公路网络中的层代表*非残差*函数。相反，我们的表述总是学习残差函数;我们的身份捷径永远不会关闭，所有信息总是被传递，同时还有额外的残差函数需要学习。此外，高速公路网络在深度极大增加（ *例如*超过 100 层）时并未显示出精度提升。

## 3 Deep Residual Learning
## 3 深度残留学习

### 3.1 Residual Learning
### 3.1 残余学习

Let us consider $\mathcal{H}(\mathbf{x})$ as an underlying mapping to be fit by a few stacked layers (not necessarily the entire net), with $\mathbf{x}$ denoting the inputs to the first of these layers. If one hypothesizes that multiple nonlinear layers can asymptotically approximate complicated functions[2]

我们假设 $\mathcal{H}(\mathbf{x})$ 为一个底层映射，由几层叠加的层拟合（不一定是整个网络），其中 $\mathbf{x}$ 表示这些层中第一个的输入。假设多个非线性层可以渐近近似复杂函数 2

, then it is equivalent to hypothesize that they can asymptotically approximate the residual functions, *i.e.*, $\mathcal{H}(\mathbf{x})-\mathbf{x}$ (assuming that the input and output are of the same dimensions). So rather than expect stacked layers to approximate $\mathcal{H}(\mathbf{x})$, we explicitly let these layers approximate a residual function $\mathcal{F}(\mathbf{x}):=\mathcal{H}(\mathbf{x})-\mathbf{x}$. The original function thus becomes $\mathcal{F}(\mathbf{x})+\mathbf{x}$. Although both forms should be able to asymptotically approximate the desired functions (as hypothesized), the ease of learning might be different.

，则等价于假设它们可以渐近近似剩余函数， *即* $\mathcal{H}(\mathbf{x})-\mathbf{x}$（假设输入和输出维数相同）。因此，我们不期望堆叠层近似 $\mathcal{H}(\mathbf{x})$，而是明确让这些层近似一个残差函数 $\mathcal{F}(\mathbf{x}):=\mathcal{H}(\mathbf{x})-\mathbf{x}$。原始函数因此变为 $\mathcal{F}(\mathbf{x})+\mathbf{x}$。虽然两种形式都应能渐近地近似所需函数（如假设），但学习难易度可能不同。

[2] This hypothesis, however, is still an open question. See [28].
然而，这一假说仍是一个悬而未决的问题。参见[28]。

This reformulation is motivated by the counterintuitive phenomena about the degradation problem (Fig. 1, left). As we discussed in the introduction, if the added layers can be constructed as identity mappings, a deeper model should have training error no greater than its shallower counterpart. The degradation problem suggests that the solvers might have difficulties in approximating identity mappings by multiple nonlinear layers. With the residual learning reformulation, if identity mappings are optimal, the solvers may simply drive the weights of the multiple nonlinear layers toward zero to approach identity mappings.

这种重新表述的动机是关于退化问题的反直觉现象（见图1，左）。正如我们在引言中讨论的，如果新增的层可以构建为身份映射，那么更深层模型的训练误差不应大于其浅层模型。退化问题表明求解者在通过多层非线性映射近似恒等映射时可能存在困难。通过残差学习重述，如果恒等映射是最优的，求解器可能只需将多层非线性层的权重推向零，以接近恒等映射。

In real cases, it is unlikely that identity mappings are optimal, but our reformulation may help to precondition the problem. If the optimal function is closer to an identity mapping than to a zero mapping, it should be easier for the solver to find the perturbations with reference to an identity mapping, than to learn the function as a new one. We show by experiments (Fig. 7) that the learned residual functions in general have small responses, suggesting that identity mappings provide reasonable preconditioning.

在实际情况中，身份映射不太可能是最优的，但我们的重新表述可能有助于预先处理这个问题。如果最优函数更接近恒等映射而非零映射，求解器应更容易找到相对于恒等映射的扰动，而不是将该函数作为新函数学习。我们通过实验展示（图。7）学习到的残差函数通常响应较小，表明恒等映射提供了合理的预条件。

### 3.2 Identity Mapping by Shortcuts
### 3.2 通过快捷方式进行身份映射

We adopt residual learning to every few stacked layers. A building block is shown in Fig. 2. Formally, in this paper we consider a building block defined as:

我们每隔几层叠加层就采用残差学习。图中展示了一个建筑模块。2. 本文形式上，我们考虑一个构建单元定义为：

$$\mathbf{y}=\mathcal{F}(\mathbf{x},\{W_{i}\})+\mathbf{x}. \tag{1}$$

Here $\mathbf{x}$ and $\mathbf{y}$ are the input and output vectors of the layers considered. The function $\mathcal{F}(\mathbf{x},\{W_{i}\})$ represents the residual mapping to be learned. For the example in Fig. 2 that has two layers, $\mathcal{F}=W_{2}\sigma(W_{1}\mathbf{x})$ in which $\sigma$ denotes ReLU [29] and the biases are omitted for simplifying notations. The operation $\mathcal{F}+\mathbf{x}$ is performed by a shortcut connection and element-wise addition. We adopt the second nonlinearity after the addition (*i.e.*, $\sigma(\mathbf{y})$, see Fig. 2).

这里 $\mathbf{x}$ 和 $\mathbf{y}$ 是所考虑图层的输入和输出向量。函数 $\mathcal{F}(\mathbf{x},\{W_{i}\})$ 表示待学习的残差映射。以图 2 有两个层，$\mathcal{F}=W_{2}\sigma(W_{1}\mathbf{x})$ 其中 $\sigma$ 表示 ReLU [29]，为了简化记号，省略了偏差。该操作 $\mathcal{F}+\mathbf{x}$ 通过快捷连接和逐元素加法实现。我们在加法后采用第二种非线性（ *即* $\sigma(\mathbf{y})$，见图）。2).

The shortcut connections in Eqn.(1) introduce neither extra parameter nor computation complexity. This is not only attractive in practice but also important in our comparisons between plain and residual networks. We can fairly compare plain/residual networks that simultaneously have the same number of parameters, depth, width, and computational cost (except for the negligible element-wise addition).

式（1）中的捷径连接既不增加额外参数，也不增加计算复杂度。这不仅在实践中具有吸引力，而且在我们比较普通网络与残差网络时也很重要。我们可以公平地比较同时拥有相同参数数、深度、宽度和计算成本的普通/残差网络（除了可忽略的元素加法）。

The dimensions of $\mathbf{x}$ and $\mathcal{F}$ must be equal in Eqn.(1). If this is not the case (*e.g.*, when changing the input/output channels), we can perform a linear projection $W_{s}$ by the shortcut connections to match the dimensions:

在方程式（1）中，和 $\mathcal{F}$ 的 $\mathbf{x}$ 维数必须相等。如果不是这样（ *例如*在更换输入/输出通道时），我们可以通过快捷连接进行线性投影 $W_{s}$，以匹配尺寸：

$$\mathbf{y}=\mathcal{F}(\mathbf{x},\{W_{i}\})+W_{s}\mathbf{x}. \tag{2}$$

We can also use a square matrix $W_{s}$ in Eqn.(1). But we will show by experiments that the identity mapping is sufficient for addressing the degradation problem and is economical, and thus $W_{s}$ is only used when matching dimensions.

我们也可以在方程式（1）中使用方阵 $W_{s}$。但我们将通过实验证明，恒等映射足以解决退化问题且经济，因此 $W_{s}$ 仅在匹配维度时使用。

The form of the residual function $\mathcal{F}$ is flexible. Experiments in this paper involve a function $\mathcal{F}$ that has two or three layers (Fig. 5), while more layers are possible. But if $\mathcal{F}$ has only a single layer, Eqn.(1) is similar to a linear layer: $\mathbf{y}=W_{1}\mathbf{x}+\mathbf{x}$, for which we have not observed advantages.

残差函数 $\mathcal{F}$ 的形式是灵活的。本文中的实验涉及一个由两层或三层组成的函数 $\mathcal{F}$（见图）。5），而多层也是可能的。但如果 $\mathcal{F}$ 只有单层，式（1）类似于线性层： $\mathbf{y}=W_{1}\mathbf{x}+\mathbf{x}$，我们尚未观察到优势。

We also note that although the above notations are about fully-connected layers for simplicity, they are applicable to convolutional layers. The function $\mathcal{F}(\mathbf{x},\{W_{i}\})$ can represent multiple convolutional layers. The element-wise addition is performed on two feature maps, channel by channel.

我们还注意到，虽然上述符号为简化起见是关于全连接层的，但它们同样适用于卷积层。该函数 $\mathcal{F}(\mathbf{x},\{W_{i}\})$ 可以表示多个卷积层。元素级的加法是在两个特征图上逐通道进行的。

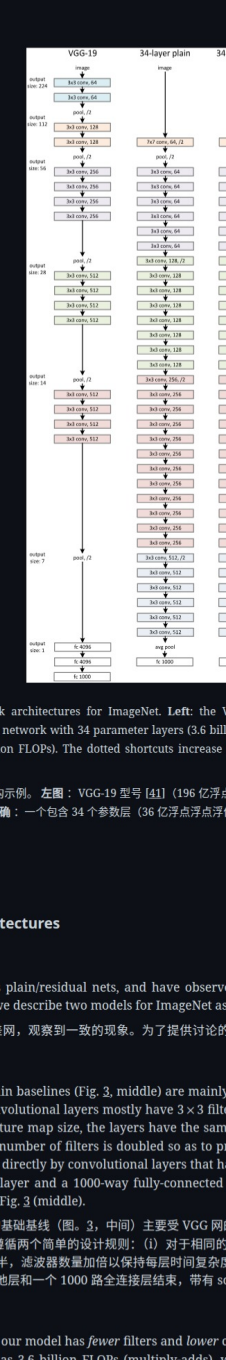


Figure 3: Example network architectures for ImageNet. **Left**: the VGG-19 model [41] (19.6 billion FLOPs) as a reference. **Middle**: a plain network with 34 parameter layers (3.6 billion FLOPs). **Right**: a residual network with 34 parameter layers (3.6 billion FLOPs). The dotted shortcuts increase dimensions. **Table 1** shows more details and other variants.

图 3： ImageNet 的网络架构示例。 **左图**： VGG-19 型号 [41]（196 亿浮点浮点）作为参考。 **中间**：一个拥有 34 个参数层（36 亿浮点）的纯网络。 **正确**：一个包含 34 个参数层（36 亿浮点浮点浮点）的残差网络。虚线快捷键增加了尺寸。 **表 1** 展示了更多细节及其他变体。

### 3.3 Network Architectures
### 3.3 网络架构

We have tested various plain/residual nets, and have observed consistent phenomena. To provide instances for discussion, we describe two models for ImageNet as follows.

我们测试了各种普通/残差网，观察到一致的现象。为了提供讨论的实例，我们描述了 ImageNet 的两个模型如下。

**Plain Network.** Our plain baselines (Fig. 3, middle) are mainly inspired by the philosophy of VGG nets [41] (Fig. 3, left). The convolutional layers mostly have 3×3 filters and follow two simple design rules: (i) for the same output feature map size, the layers have the same number of filters; and (ii) if the feature map size is halved, the number of filters is doubled so as to preserve the time complexity per layer. We perform downsampling directly by convolutional layers that have a stride of 2. The network ends with a global average pooling layer and a 1000-way fully-connected layer with softmax. The total number of weighted layers is 34 in Fig. 3 (middle).

**Plain Network。** 我们的基础基线（图。3，中间）主要受 VGG 网的哲学[41] 启发（见图。3，左）。卷积层大多有 3 × 3 滤波器，并遵循两个简单的设计规则：（i）对于相同的输出特征映射大小，各层的滤波器数量相同;（ii）如果特征图大小减半，滤波器数量加倍以保持每层时间复杂度。我们直接通过步幅为 2 的卷积层进行降采样。网络以一个全球平均池层和一个 1000 路全连接层结束，带有 softmax。加重层总数为 34 层（见图）。3（中间）。

It is worth noticing that our model has *fewer* filters and *lower* complexity than VGG nets [41] (Fig. 3, left). Our 34-layer baseline has 3.6 billion FLOPs (multiply-adds), which is only 18% of VGG-19 (19.6 billion FLOPs).

值得注意的是，我们的模型比 VGG 网的*滤波器更少*，复杂*度更低* [41]（见图。3，左）。我们的 34 层基线有 36 亿次 FLOP（乘法加法），仅占 VGG-19（196 亿次 FLOP）的 18%。

**Residual Network.** Based on the above plain network, we insert shortcut connections (Fig. 3, right) which turn the network into its counterpart residual version. The identity shortcuts (Eqn.(1)) can be directly used when the input and output are of the same dimensions (solid line shortcuts in Fig. 3). When the dimensions increase (dotted line shortcuts in Fig. 3), we consider two options: (A) The shortcut still performs identity mapping, with extra zero entries padded for increasing dimensions. This option introduces no extra parameter; (B) The projection shortcut in Eqn.(2) is used to match dimensions (done by 1×1 convolutions). For both options, when the shortcuts go across feature maps of two sizes, they are performed with a stride of 2.

**残留网络。** 基于上述纯网络，我们插入快捷连接（见图。3，右侧），将网络转换为其对应的残差版本。当输入和输出尺寸相同时，恒定捷径（式（1））可以直接使用（图中的实线捷径）。3）当尺寸增加时（图中的虚线捷径）。3），我们考虑两种选项：（A） 快捷方式仍执行恒等映射，并根据维数增加填充额外的零条目。该选项不引入额外参数;（B） 方程（2）中的投影捷径用于匹配维度（通过 1×1 卷积实现）。对于这两个选项，当快捷键跨越两个尺寸的特征图时，它们的步幅是 2。

| layer name 图层名称 | output size 输出尺寸 | 18-layer 18层 | 34-layer 34层 | 50-layer 50层 | 101-layer 101层 | 152-layer 152层 |
|---|---|---|---|---|---|---|
| conv1 | 112×112 | 7×7, 64, stride 2<br>7 × 7,64，步幅 2 | | | | |
| conv2_x | 56×56 | 3×3 max pool, stride 2<br>3 × 3 最大池化，步幅 2 | | | | |
| | | $\left[\begin{array}{c}3\times3,64\\3\times3,64\end{array}\right]\times2$ | $\left[\begin{array}{c}3\times3,64\\3\times3,64\end{array}\right]\times3$ | $\left[\begin{array}{c}1\times1,64\\3\times3,64\\1\times1,256\end{array}\right]\times3$ | $\left[\begin{array}{c}1\times1,64\\3\times3,64\\1\times1,256\end{array}\right]\times3$ | $\left[\begin{array}{c}1\times1,64\\3\times3,64\\1\times1,256\end{array}\right]\times3$ |
| conv3_x | 28×28 | $\left[\begin{array}{c}3\times3,128\\3\times3,128\end{array}\right]\times2$ | $\left[\begin{array}{c}3\times3,128\\3\times3,128\end{array}\right]\times4$ | $\left[\begin{array}{c}1\times1,128\\3\times3,128\\1\times1,512\end{array}\right]\times4$ | $\left[\begin{array}{c}1\times1,128\\3\times3,128\\1\times1,512\end{array}\right]\times4$ | $\left[\begin{array}{c}1\times1,128\\3\times3,128\\1\times1,512\end{array}\right]\times8$ |
| conv4_x | 14×14 | $\left[\begin{array}{c}3\times3,256\\3\times3,256\end{array}\right]\times2$ | $\left[\begin{array}{c}3\times3,256\\3\times3,256\end{array}\right]\times6$ | $\left[\begin{array}{c}1\times1,256\\3\times3,256\\1\times1,1024\end{array}\right]\times6$ | $\left[\begin{array}{c}1\times1,256\\3\times3,256\\1\times1,1024\end{array}\right]\times23$ | $\left[\begin{array}{c}1\times1,256\\3\times3,256\\1\times1,1024\end{array}\right]\times36$ |
| conv5_x | 7×7 | $\left[\begin{array}{c}3\times3,512\\3\times3,512\end{array}\right]\times2$ | $\left[\begin{array}{c}3\times3,512\\3\times3,512\end{array}\right]\times3$ | $\left[\begin{array}{c}1\times1,512\\3\times3,512\\1\times1,2048\end{array}\right]\times3$ | $\left[\begin{array}{c}1\times1,512\\3\times3,512\\1\times1,2048\end{array}\right]\times3$ | $\left[\begin{array}{c}1\times1,512\\3\times3,512\\1\times1,2048\end{array}\right]\times3$ |
| | 1×1 | average pool, 1000-d fc, softmax<br>平均池，1000-D FC，Softmax | | | | |
| FLOPs 翻牌 | | $1.8\times10^{9}$ | $3.6\times10^{9}$ | $3.8\times10^{9}$ | $7.6\times10^{9}$ | $11.3\times10^{9}$ |

Table 1: Architectures for ImageNet. Building blocks are shown in brackets (see also Fig. 5), with the numbers of blocks stacked. Downsampling is performed by conv3_1, conv4_1, and conv5_1 with a stride of 2.

表 1： ImageNet 架构。构建模块以括号示（另见图）。5），并叠加了方块数量。降采样由 conv3_1、conv4_1 和 conv5_1 以 2 步幅进行。

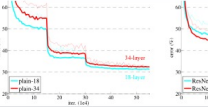


Figure 4: Training on **ImageNet**. Thin curves denote training error, and bold curves denote validation error of the center crops. Left: plain networks of 18 and 34 layers. Right: ResNets of 18 and 34 layers. In this plot, the residual networks have no extra parameter compared to their plain counterparts.

图 4： **ImageNet** 上的训练。细曲线表示训练误差，粗体曲线表示中心作物的验证误差。左图：18 层和 34 层的普通网络。右图：18 层和 34 层的 ResNet。在该图中，残差网络相比普通网络没有额外的参数。

### 3.4 Implementation
### 3.4 实现

Our implementation for ImageNet follows the practice in [21, 41]. The image is resized with its shorter side randomly sampled in $[256,480]$ for scale augmentation [41]. A 224×224 crop is randomly sampled from an image or its horizontal flip, with the per-pixel mean subtracted [21]. The standard color augmentation in [21] is used. We adopt batch normalization (BN) [16] right after each convolution and before activation, following [16]. We initialize the weights as in [13] and train all plain/residual nets from scratch. We use SGD with a mini-batch size of 256. The learning rate starts from 0.1 and is divided by 10 when the error plateaus, and the models are trained for up to $60\times10^{4}$ iterations. We use a weight decay of 0.0001 and a momentum of 0.9. We do not use dropout [14], following the practice in [16].

我们对 ImageNet 的实现遵循了[21， 41] 中的做法。图像被重新调整大小，短边随机采样 $[256,480]$ 以进行比例放大 [41]。从图像或其水平翻转中随机采样 224 × 224 裁剪，减去每像素平均值 [21]。采用了[21] 中的标准色彩增强。我们在每次卷积后、激活前，之后[16] 采用批次归一化（BN） [16]。我们像 [13] 那样初始化权重，并从零训练所有普通/残差网络。我们使用 SGD，小批量为 256。学习率从 0.1 开始，误差停滞时除以 10，模型可训练最多可达 $60\times10^{4}$ 重复次数。我们使用权重衰减 0.0001，动量 0.9。我们不使用退出 [14]，遵循 [16] 中的做法。

In testing, for comparison studies we adopt the standard 10-crop testing [21]. For best results, we adopt the fully-convolutional form as in [41, 13], and average the scores at multiple scales (images are resized such that the shorter side is in $\{224,256,384,480,640\}$).

在测试中，比较研究采用标准的 10 作物测试 [21]。为了获得最佳结果，我们采用如 [41， 13] 中的全卷积形式，并在多个尺度上平均分数（图像缩小使短边在 $\{224,256,384,480,640\}$）。

## 4 Experiments 4 实验

### 4.1 ImageNet Classification
### 4.1 ImageNet 分类

We evaluate our method on the ImageNet 2012 classification dataset [36] that consists of 1000 classes. The models are trained on the 1.28 million training images, and evaluated on the 50k validation images. We also obtain a final result on the 100k test images, reported by the test server. We evaluate both top-1 and top-5 error rates.

我们在包含 1000 个类别的 ImageNet 2012 分类数据集 [36] 上评估了我们的方法。模型基于 128 万张训练图像进行训练，并在 5 万张验证图像上进行评估。我们还会获得测试服务器报告的 10 万张测试图像的最终结果。我们评估了前一和前五的错误率。

**Plain Networks.** We first evaluate 18-layer and 34-layer plain nets. The 34-layer plain net is in Fig. 3 (middle). The 18-layer plain net is of a similar form. See Table 1 for detailed architectures.

**纯网络。** 我们首先评估 18 层和 34 层平网。34 层的平网如图所示。3（中间）。18 层的平网形式类似。详见表 1 了解详细架构。

The results in Table 2 show that the deeper 34-layer plain net has higher validation error than the shallower 18-layer plain net. To reveal the reasons, in Fig. 4 (left) we compare their training/validation errors during the training procedure. We have observed the degradation problem - the 34-layer plain net has higher *training* error throughout the whole training procedure, even though the solution space of the 18-layer plain network is a subspace of that of the 34-layer one.

表 2 的结果显示，更深的 34 层平网的验证误差高于较浅的 18 层平网。为揭示原因，见图。4（左）我们比较它们在训练过程中的训练/验证误差。我们观察到退化问题——34 层平面网在整个训练过程中*训练误差*较高，尽管 18 层平面网络的解空间是 34 层网络的子空间。

| | plain 平原 | ResNet |
|---|---|---|
| 18 layers 18层 | 27.94 | 27.88 |
| 34 layers 34层 | 28.54 | **25.03** |

Table 2: Top-1 error (%, 10-crop testing) on ImageNet validation. Here the ResNets have no extra parameter compared to their plain counterparts. Fig. 4 shows the training procedures.

表 2： ImageNet 验证中的前一误差（百分比，10 次截图测试）。这里的 ResNet 与普通的版本相比没有额外参数。无花果。4 展示了训练程序。

We argue that this optimization difficulty is *unlikely* to be caused by vanishing gradients. These plain networks are trained with BN [16], which ensures forward propagated signals to have non-zero variances. We also verify that the backward propagated gradients exhibit healthy norms with BN. So neither forward nor backward signals vanish. In fact, the 34-layer plain net is still able to achieve competitive accuracy (Table 3), suggesting that the solver works to some extent. We conjecture that the deep plain nets may have exponentially low convergence rates, which impact the reducing of the training error[3]
我们认为，这种优化难度不太*可能*是梯度消失引起的。这些平面网络采用 BN [16] 训练，确保前向传播信号具有非零方差。我们还验证了反向传播梯度在 BN 下表现出健康的范数。所以正向和反向信号都不会消失。事实上，34 层的平网依然能够实现竞争性精度（见表 3），表明求解器在一定程度上是有效的。我们推测深平网的收敛率可能呈指数级下降，这会影响训练误差的减少 3. The reason for such optimization difficulties will be studied in the future.
.这些优化难度的原因将在未来研究。

[3] We have experimented with more training iterations (3×) and still observed the degradation problem, suggesting that this problem cannot be feasibly addressed by simply using more iterations.
我们尝试了更多的训练迭代（3 次 ×），仍然观察到退化问题，表明仅仅通过增加迭代次数无法可行地解决这一问题。

**Residual Networks.** Next we evaluate 18-layer and 34-layer residual nets (*ResNets*). The baseline architectures are the same as the above plain nets, expect that a shortcut connection is added to each pair of 3×3 filters as in Fig. 3 (right). In the first comparison (Table 2 and Fig. 4 right), we use identity mapping for all shortcuts and zero-padding for increasing dimensions (option A). So they have *no extra parameter* compared to the plain counterparts.

**残留网络。** 接下来我们评估 18 层和 34 层残差网（*ResNets*）。基线架构与上述纯网相同，预计每对 3 × 3 个滤波器都添加了快捷连接，如图所示。3（右）。在第一次比较中（表 2 和图）。4 右），我们对所有快捷方式使用恒等映射，对尺寸增加使用零填充（选项 A）。所以与普通版本相比，它们*没有额外参数*。

We have three major observations from Table 2 and Fig. 4. First, the situation is reversed with residual learning – the 34-layer ResNet is better than the 18-layer ResNet (by 2.8%). More importantly, the 34-layer ResNet exhibits considerably lower training error and is generalizable to the validation data. This indicates that the degradation problem is well addressed in this setting and we manage to obtain accuracy gains from increased depth.

我们从表 2 和图中得出三个主要观察。4. 首先，残差学习情况则相反——34 层 ResNet 优于 18 层 ResNet（高出 2.8%）。更重要的是，34 层 ResNet 的训练误差显著更低，并且可推广到验证数据。这表明在此情况下退化问题得到了很好的解决，我们也成功地通过增加深度获得了精度提升。

Second, compared to its plain counterpart, the 34-layer ResNet reduces the top-1 error by 3.5% (Table 2), resulting from the successfully reduced training error (Fig. 4 right *vs.* left). This comparison verifies the effectiveness of residual learning on extremely deep systems.

其次，与普通版本相比，34 层 ResNet 将前一误差降低了 3.5%（见表 2），这得益于成功降低了训练误差（图。4 右 *vs.* 左）。这种比较验证了残差学习在极深系统上的有效性。

Last, we also note that the 18-layer plain/residual nets are comparably accurate (Table 2), but the 18-layer ResNet converges faster (Fig. 4 right *vs.* left). When the net is “not overly deep” (18 layers here), the current SGD solver is still able to find good solutions to the plain net. In this case, the ResNet eases the optimization by providing faster convergence at the early stage.

最后，我们还注意到 18 层平面/残差网的准确度相当（见表 2），但 18 层的 ResNet 收敛速度更快（见图。4 右 *vs.* 左）。当网络“不算过深”（这里是 18 层）时，当前的 SGD 求解器仍能为普通网络找到良好的解。在这种情况下，ResNet 通过在早期阶段提供更快的收敛，简化了优化。

| model | top-1 err. 前一，错。 | top-5 err. 前五名，错。 |
|---|---|---|
| VGG-16 [41] | 28.07 | 9.33 |
| GoogLeNet [44]<br>谷歌 LeNet [44] | - | 9.15 |
| PReLU-net [13]<br>PReLU 网络 [13] | 24.27 | 7.38 |
| plain-34 普通34 | 28.54 | 10.02 |
| ResNet-34 A | 25.03 | 7.76 |
| ResNet-34 B | 24.52 | 7.46 |
| ResNet-34 C | 24.19 | 7.40 |
| ResNet-50 | 22.85 | 6.71 |
| ResNet-101 | 21.75 | 6.05 |
| ResNet-152 | **21.43** | **5.71** |

Table 3: Error rates (%, **10-crop** testing) on ImageNet validation. VGG-16 is based on our test. ResNet-50/101/152 are of option B that only uses projections for increasing dimensions.

表 3： ImageNet 验证中的错误率（百分比，**10 次裁剪测试**）。VGG-16 是基于我们的测试。ResNet-50/101/152 属于选项 B，只使用投影来实现维度递增。

| method 方法 | top-1 err. 前一，错。 | top-5 err. 前五名，错。 |
|---|---|---|
| VGG [41] (ILSVRC'14)<br>VGG [41] (ILSVRC'14) | - | 8.43† |
| GoogLeNet [44] (ILSVRC'14)<br>GoogLeNet [44] (ILSVRC'14) | - | 7.89 |
| VGG [41] (v5)<br>VGG [41] (v5) | 24.4 | 7.1 |
| PReLU-net [13]<br>PReLU 网络 [13] | 21.59 | 5.71 |
| BN-inception [16]<br>BN 的诞生 [16] | 21.99 | 5.81 |
| ResNet-34 B | 21.84 | 5.71 |
| ResNet-34 C | 21.53 | 5.60 |
| ResNet-50 | 20.74 | 5.25 |
| ResNet-101 | 19.87 | 4.60 |
| ResNet-152 | **19.38** | **4.49** |

Table 4: Error rates (%) of **single-model** results on the ImageNet validation set (except † reported on the test set).

表 4： ImageNet 验证集单**模型**结果的错误率（百分比）（测试集报告的 † 除外）。

| method 方法 | top-5 err. (test) 前五名，错。（测试） |
|---|---|
| VGG [41] (ILSVRC'14)<br>VGG [41] (ILSVRC'14) | 7.32 |
| GoogLeNet [44] (ILSVRC'14)<br>GoogLeNet [44] (ILSVRC'14) | 6.66 |
| VGG [41] (v5)<br>VGG [41] (v5) | 6.8 |
| PReLU-net [13]<br>PReLU 网络 [13] | 4.94 |
| BN-inception [16]<br>BN 的诞生 [16] | 4.82 |
| **ResNet (ILSVRC'15) ResNet (ILSVRC'15)** | **3.57** |

Table 5: Error rates (%) of **ensembles**. The top-5 error is on the test set of ImageNet and reported by the test server.

表 5： **合奏**的错误率（%）。前五个错误出现在 ImageNet 测试集，由测试服务器报告。

**Identity vs. Projection Shortcuts.** We have shown that parameter-free, identity shortcuts help with training. Next we investigate projection shortcuts (Eqn.(2)). In Table 3 we compare three options: (A) zero-padding shortcuts are used for increasing dimensions, and all shortcuts are parameter-free (the same as Table 2 and Fig. 4 right); (B) projection shortcuts are used for increasing dimensions, and other shortcuts are identity; and (C) all shortcuts are projections.

**身份 vs 投影捷径。** 我们已经证明，无参数的身份捷径有助于训练。接下来我们研究投影捷径（式（2））。在表 3 中，我们比较了三种选项：（A） 用于增加维度的零填充捷径，且所有捷径均无参数（与表 2 和图相同）。4 右）;（B） 投影捷径用于增加维度，其他捷径为恒等;以及（C）所有捷径都是投影。

Table 3 shows that all three options are considerably better than the plain counterpart. B is slightly better than A. We argue that this is because the zero-padded dimensions in A indeed have no residual learning. C is marginally better than B, and we attribute this to the extra parameters introduced by many (thirteen) projection shortcuts. But the small differences among A/B/C indicate that projection shortcuts are not essential for addressing the degradation problem. So we do not use option C in the rest of this paper, to reduce memory/time complexity and model sizes. Identity shortcuts are particularly important for not increasing the complexity of the bottleneck architectures that are introduced below.

表 3 显示，这三种选项都明显优于普通选项。B 比 A 稍微好一点。我们认为这是因为 A 中零填充维度确实没有残余学习。C 比 B 略好一些，我们将此归因于许多（十三个）投影捷径引入的额外参数。但 A/B/C 之间的细微差异表明，投影捷径并非解决退化问题的必要手段。因此，本文其余部分不使用选项 C，以降低内存/时间复杂度和模型大小。身份捷径对于不增加下面介绍的瓶颈架构复杂性尤为重要。

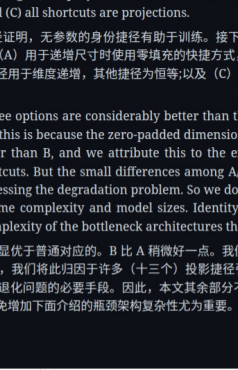


Figure 5: A deeper residual function $\mathcal{F}$ for ImageNet. Left: a building block (on 56×56 feature maps) as in Fig. 3 for ResNet-34. Right: a “bottleneck” building block for ResNet-50/101/152.

图 5： ImageNet 的更深残差函数 \mathcal{F}。左图：构建模块（在 56 × 56 张特征图上），如图所示。3 为 ResNet-34。右图：ResNet-50/101/152 的“瓶颈”构建模块。

**Deeper Bottleneck Architectures.** Next we describe our deeper nets for ImageNet. Because of concerns on the training time that we can afford, we modify the building block as a *bottleneck* design[4]
**更深层的瓶颈架构。** 接下来我们介绍 ImageNet 的更深层网络。由于对训练时间的担忧，我们将构建模块修改为*瓶颈*设计 4. For each residual function $\mathcal{F}$, we use a stack of 3 layers instead of 2 (Fig. 5). The three layers are 1×1, 3×3, and 1×1 convolutions, where the 1×1 layers are responsible for reducing and then increasing (restoring) dimensions, leaving the 3×3 layer a bottleneck with smaller input/output dimensions. Fig. 5 shows an example, where both designs have similar time complexity.
.对于每个残差函数 $\mathcal{F}$，我们使用三层叠加而非两层（见图）。5）。这三层分别是 1 × 1、3 × 3 和 1 × 1 卷积，其中 1 × 1 层负责减少再增加（恢复）维度，使 3 × 3 层成为输入/输出维度较小的瓶颈。无花果。5 展示了一个例子，两个设计的时间复杂度都相似。

[4] Deeper non-bottleneck ResNets (*e.g.*, Fig. 5 left) also gain accuracy from increased depth (as shown on CIFAR-10), but are not as economical as the bottleneck ResNets. So the usage of bottleneck designs is mainly due to practical considerations. We further note that the degradation problem of plain nets is also witnessed for the bottleneck designs.
更深层的非瓶颈 ResNet（*例如*，图。5 左）也从深度增加中获得了准确性（如 CIFAR-10 所示），但不如瓶颈 ResNet 经济实惠。因此，瓶颈设计的使用主要出于实际考虑。我们进一步注意到，瓶颈设计中也存在普通网的退化问题。

The parameter-free identity shortcuts are particularly important for the bottleneck architectures. If the identity shortcut in Fig. 5 (right) is replaced with projection, one can show that the time complexity and model size are doubled, as the shortcut is connected to the two high-dimensional ends. So identity shortcuts lead to more efficient models for the bottleneck designs.

无参数的恒等捷径对于瓶颈架构尤为重要。如果图中的身份捷径。5（右）被投影替代，可以看出时间复杂度和模型大小加倍，因为捷径连接到两个高维端。因此，身份捷径为瓶颈设计带来了更高效的模型。

**50-layer ResNet:** We replace each 2-layer block in the 34-layer net with this 3-layer bottleneck block, resulting in a 50-layer ResNet (Table 1). We use option B for increasing dimensions. This model has 3.8 billion FLOPs.

**50 层 ResNet：** 我们将 34 层网络中的每个两层块替换为这个三层瓶颈块，形成一个 50 层的 ResNet（见表 1）。我们使用选项 B 来增加维度。该模型拥有 38 亿次浮点运算。

**101-layer and 152-layer ResNets:** We construct 101-layer and 152-layer ResNets by using more 3-layer blocks (Table 1). Remarkably, although the depth is significantly increased, the 152-layer ResNet (11.3 billion FLOPs) still has *lower complexity* than VGG-16/19 nets (15.3/19.6 billion FLOPs).

**101 层和 152 层 ResNet：** 我们通过使用更多的三层块构建 101 层和 152 层的 ResNet（见表 1）。值得注意的是，尽管深度显著增加，152 层的 ResNet（113 亿次浮点浮点）的*复杂度*仍低于 VGG-16/19 网络（153 亿/196 亿次浮点）。

The 50/101/152-layer ResNets are more accurate than the 34-layer ones by considerable margins (Table 3 and 5). We do not observe the degradation problem and thus enjoy significant accuracy gains from considerably increased depth. The benefits of depth are witnessed for all evaluation metrics (Table 3 and 5).

50/101/152 层的 ResNet 比 34 层的更准确（见表 3 和表 5）。我们不观察退化问题，因此能够从深度大幅提升中获得显著的准确性提升。深度的好处在所有评估指标中都可见（见表 3 和表 5）。

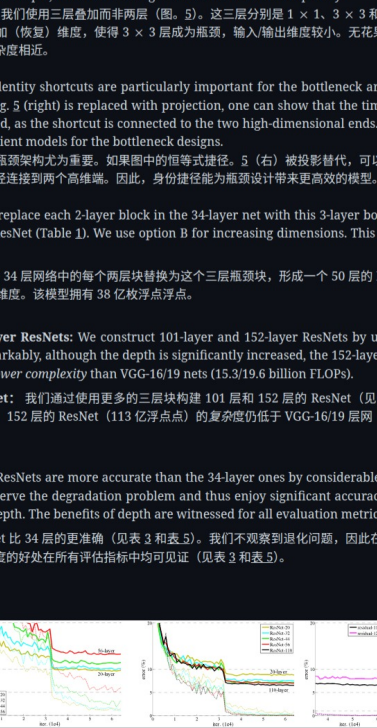


Figure 6: Training on **CIFAR-10**. Dashed lines denote training error, and bold lines denote testing error. Left: plain networks. The error of plain-110 is higher than 60% and not displayed. Middle: ResNets. Right: ResNets with 110 and 1202 layers.

图 6： **CIFAR-10** 训练。虚线表示训练误差，粗线表示测试误差。 **左**：普通网络。平原 110 的误差高于 60%，但未显示。 **中间**：ResNets。 **右图**：110 和 1202 层的 ResNet。

**Comparisons with State-of-the-art Methods.** In Table 5 we compare with the previous best single-model results. Our baseline 34-layer ResNets have achieved very competitive accuracy. Our 152-layer ResNet has a single-model top-5 validation error of 4.49%. This single-model result outperforms all previous ensemble results (Table 5). We combine six models of different depth to form an ensemble (only with two 152-layer ones at the time of submitting). This leads to **3.57%** top-5 error on the test set (Table 5). *This entry won the 1st place in ILSVRC 2015.*

**与最先进方法的比较。** 在表 5 中，我们与之前的最佳单模型结果进行比较。我们的基础 34 层 ResNet 实现了极具竞争力的准确性。我们的 152 层 ResNet 单模型前五验证误差为 4.49%。该单一模型结果优于之前所有合奏结果（见表 5）。我们结合了六个不同深度的模型组成一个集合（提交时仅有两个 152 层模型）。这导致测试集的前五误差为 **3.57%**（见表 5）。 *该作品获得了 2015 年 ILSVRC 比赛中获得第一名。*

### 4.2 CIFAR-10 and Analysis
### 4.2 CIFAR-10 与分析

We conducted more studies on the CIFAR-10 dataset [20], which consists of 50k training images and 10k testing images in 10 classes. We present experiments trained on the training set and evaluated on the test set. Our focus is on the behaviors of extremely deep networks, but not on pushing the state-of-the-art results, so we intentionally use simple architectures as follows.

我们对 CIFAR-10 数据集[20] 进行了更多研究，该数据集包含 10 门课的 5 万训练图像和 1 万张测试图像。我们展示在训练集上训练并在测试集上评估的实验。我们关注的是极深网络的行为，但不是推动最先进的结果，因此我们有意使用以下简单的架构。

The plain/residual architectures follow the form in Fig. 3 (middle/right). The network inputs are 32×32 images, with the per-pixel mean subtracted. The first layer is 3×3 convolutions. Then we use a stack of $6n$ layers with 3×3 convolutions on the feature maps of sizes $\{32,16,8\}$ respectively, with $2n$ layers for each feature map size. The numbers of filters are $\{16,32,64\}$ respectively. The subsampling is performed by convolutions with a stride of 2. The network ends with a global average pooling, a 10-way fully-connected layer, and softmax. There are totally $6n+2$ stacked weighted layers. The following table summarizes the architecture:

平原/残余建筑遵循图中的形式。3（中/右）。网络输入为 32 × 32 张图像，每像素平均值被减去。第一层是 3 × 3 个卷积。然后我们使用 $6n$ 一组图层，每个 × 特征图 $\{32,16,8\}$ 大小分别对应 3 个卷积，每个特征图大小各有 $2n$ 个图层。滤波器的数量分别是 $\{16,32,64\}$。子采样通过步幅为 2 的卷积进行。网络以全球平均池化、10 路全连接层和软最大值结束。共有 6 层 $n+2$ 叠加重量层。下表总结了该架构：

| output map size 输出映射大小 | 32×32 | 16×16 | 8×8 |
|---|---|---|---|
| # layers # 层数 | 1+2$n$ | 2$n$ | 2$n$ |
| # filters # 滤镜 | 16 | 32 | 64 |

When shortcut connections are used, they are connected to the pairs of 3×3 layers (totally $3n$ shortcuts). On this dataset we use identity shortcuts in all cases (*i.e.*, option A), so our residual models have exactly the same depth, width, and number of parameters as the plain counterparts.

使用快捷连接时，它们连接到三 × 层的对（ $3n$ 总捷径）。在这个数据集上，我们在所有情况下都使用身份捷径（ *即*选项 A），因此我们的残差模型的深度、宽度和参数数与普通模型完全相同。

| method 方法 | | | error (%) 误差（%） |
|---|---|---|---|
| Maxout [10]<br>极限 [10] | | | 9.38 |
| NIN [25]<br>[25] | | | 8.81 |
| DSN [24] | | | 8.22 |
| | # layers # 层数 | # params # 参数 | |
| FitNet [35] | 19 | 2.5M | 8.39 |
| Highway [42, 43]<br>公路 [42, 43] | 19 | 2.3M | 7.54 (7.72±0.16)<br>7.54 (7.72±\pm0.16) |
| Highway [42, 43]<br>公路 [42, 43] | 32 | 1.25M | 8.80 |
| ResNet | 20 | 0.27M | 8.75 |
| ResNet | 32 | 0.46M | 7.51 |
| ResNet | 44 | 0.66M | 7.17 |
| ResNet | 56 | 0.85M | 6.97 |
| ResNet | 110 | 1.7M | **6.43** (6.61±0.16)<br>**6.43** (6.61±\pm0.16) |
| ResNet | 1202 | 19.4M | 7.93 |

Table 6: Classification error on the **CIFAR-10** test set. All methods are with data augmentation. For ResNet-110, we run it 5 times and show “best (mean±std)” as in [43].

表 6： CIFAR-10 测试集上的分类误差。所有方法都采用数据增强。对于 ResNet-110，我们运行 5 次，显示“最佳（平均\pmstd）” 如[43]。

We use a weight decay of 0.0001 and momentum of 0.9, and adopt the weight initialization in [13] and BN [16] but with no dropout. These models are trained with a mini-batch size of 128 on two GPUs. We start with a learning rate of 0.1, divide it by 10 at 32k and 48k iterations, and terminate training at 64k iterations, which is determined on a 45k/5k train/val split. We follow the simple data augmentation in [24] for training: 4 pixels are padded on each side, and a 32×32 crop is randomly sampled from the padded image or its horizontal flip. For testing, we only evaluate the single view of the original 32×32 image.

我们使用权重衰减为 0.0001，动量为 0.9，并在[13] 和 BN [16] 中采用权重初始化，但没有退出。这些模型在两块 GPU 上以 128 小批量训练。我们从 0.1 的学习率开始，32k 和 48k 迭代时除以 10，64k 迭代时终止训练，该率由 45k/5k 火车/阀门分割决定。我们按照[24] 中的简单数据增强进行训练：每侧填充 4 个像素，从填充图像或其水平翻转中随机采样 32 × 32 裁剪。测试时，我们只评估原始 32 × 32 图像的单一视图。

We compare $n=\{3,5,7,9\}$, leading to 20, 32, 44, and 56-layer networks. Fig. 6 (left) shows the behaviors of the plain nets. The deep plain nets suffer from increased depth, and exhibit higher training error when going deeper. This phenomenon is similar to that on ImageNet (Fig. 4, left) and on MNIST (see [42]), suggesting that such an optimization difficulty is a fundamental problem.

我们比较 $n=\{3,5,7,9\}$，得到 20、32、44 和 56 层网络。无花果。6（左）展示了平网的行为。深平网深度增加，深入时训练误差更高。这种现象与 ImageNet 上的类似（见图）。4，左）以及关于 MNIST（见[42]），表明这种优化难度是一个根本性问题。

Fig. 6 (middle) shows the behaviors of ResNets. Also similar to the ImageNet cases (Fig. 4, right), our ResNets manage to overcome the optimization difficulty and demonstrate accuracy gains when the depth increases.

无花果。6（中间）展示了 ResNet 的行为。也类似于 ImageNet 的案例（图。4，右），我们的 ResNet 成功克服了优化难度，并在深度增加时展示了精度提升。

We further explore $n=18$ that leads to a 110-layer ResNet. In this case, we find that the initial learning rate of 0.1 is slightly too large to start converging[5]
我们进一步探讨 $n=18$ 了导致了 110 层的 ResNet。在这种情况下，我们发现初始学习率 0.1 略大，无法开始收敛 5. So we use 0.01 to warm up the training until the training error is below 80% (about 400 iterations), and then go back to 0.1 and continue training. The rest of the learning schedule is as done previously. This 110-layer network converges well (Fig. 6, middle). It has *fewer* parameters than other deep and thin networks such as FitNet [35] and Highway [42] (Table 6), yet is among the state-of-the-art results (6.43%, Table 6).
.所以我们用 0.01 来预热训练，直到训练误差低于 80%（大约 400 次迭代），然后回到 0.1 继续训练。其余学习计划如前所述。这个 110 层网络收敛得很好（见图）。6，中间）。其参数比 FitNet [35] 和 Highway [42] 等深度细网络*更少*（见表 6），但却是最先进的结果之一（6.43%，见表 6）。

[5] With an initial learning rate of 0.1, it starts converging (<90% error) after several epochs, but still reaches similar accuracy.
初始学习率为 0.1，经过几个纪元后开始收敛（< 误差 90%），但仍达到类似的精度。

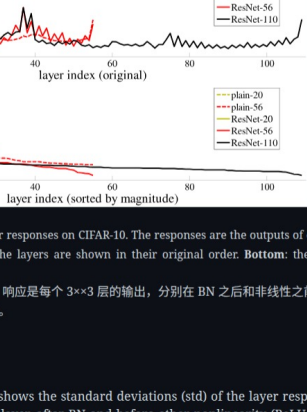


Figure 7: Standard deviations (std) of layer responses on CIFAR-10. The responses are the outputs of each 3×3 layer, after BN and before nonlinearity. **Top**: the layers are shown in their original order. **Bottom**: the responses are ranked in descending order.

图 7： CIFAR-10 层响应的标准差（std）。响应是每个 3×3 层的输出，分别在 BN 之后和非线性之前。 **顶部**：图层按原始顺序排列。 **底部**：回答按降序排列。

**Analysis of Layer Responses.** Fig. 7 shows the standard deviations (std) of the layer responses. The responses are the outputs of each 3×3 layer, after BN and before other nonlinearity (ReLU/addition). For ResNets, this analysis reveals the response strength of the residual functions. Fig. 7 shows that ResNets have generally smaller responses than their plain counterparts. These results support our basic motivation (Sec.3.1) that the residual functions might be generally closer to zero than the non-residual functions. We also notice that the deeper ResNet has smaller magnitudes of responses, as evidenced by the comparisons among ResNet-20, 56, and 110 in Fig. 7. When there are more layers, an individual layer of ResNets tends to modify the signal less.

**层响应分析。** 无花果。7 显示层响应的标准差（std）。响应是每个 3 × 3 层的输出，经过 BN 且在其他非线性（ReLU/加法）之前。对于 ResNet，该分析揭示了残差函数的响应强度。无花果。7 显示 ResNet 的响应通常比普通版本更小。这些结果支持了我们的基本动机（第 3.1 节），即残差函数可能普遍比非残差函数更接近零。我们还注意到，更深层的 ResNet 响应幅度较小，这可以从 ResNet-20、56 和 110 图中的比较中看出。7. 当层数更多时，单个 ResNet 层对信号的修改较少。

**Exploring Over 1000 layers.** We explore an aggressively deep model of over 1000 layers. We set $n=200$ that leads to a 1202-layer network, which is trained as described above. Our method shows *no optimization difficulty*, and this $10^{3}$-layer network is able to achieve *training error* <0.1% (Fig. 6, right). Its test error is still fairly good (7.93%, Table 6).

**探索超过 1000 层。** 我们探索了一个包含超过 1000 层的极深模型。我们设置 $n=200$，形成一个 1202 层网络，训练方式如上所述。我们的方法显示*无优化难度*，这个 $10^{3}$ 层网络能够实现*训练误差* < 0.1%（见图）。6，右）。其测试误差依然相当不错（7.93%，见表 6）。

But there are still open problems on such aggressively deep models. The testing result of this 1202-layer network is worse than that of our 110-layer network, although both have similar training error. We argue that this is because of overfitting. The 1202-layer network may be unnecessarily large (19.4M) for this small dataset. Strong regularization such as maxout [10] or dropout [14] is applied to obtain the best results ([10, 25, 24, 35]) on this dataset. In this paper, we use no maxout/dropout and just simply impose regularization via deep and thin architectures by design, without distracting from the focus on the difficulties of optimization. But combining with stronger regularization may improve results, which we will study in the future.

但对于如此深度的深度模型，仍存在一些未解之谜。这个 1202 层网络的测试结果比我们 110 层网络差，尽管两者的训练误差相似。我们认为这是因为过拟合。对于这个小数据集来说，1202 层网络可能不必要地大（1940 万）。采用强正则化如 maxout [10] 或 dropout [14] 以获得该数据集的最佳结果（[10， 25， 24， 35]）。本文中，我们不使用最大/退出，而是通过设计中深层和细致架构来强制正则化，而不偏离对优化难点的关注。但结合更强的正则化可能会改善结果，我们未来将对此进行研究。

### 4.3 Object Detection on PASCAL and MS COCO
### 4.3 PASCAL 和 MS COCO 上的物体探测

| training data 训练数据 | 07+12 | 07++12 |
|---|---|---|
| test data 测试数据 | VOC 07 test VOC 07 测试 | VOC 12 test VOC 12 测试 |
| VGG-16 | 73.2 | 70.4 |
| ResNet-101 | **76.4** | **73.8** |

Table 7: Object detection mAP (%) on the PASCAL VOC 2007/2012 test sets using **baseline** Faster R-CNN. See also Table 10 and 11 for better results.

表 7： 基于基线 Faster R-CNN 的 PASCAL VOC 2007/2012 测试集对象检测 mAP（%）。另见表 11 和表 11 以获得更好的结果。

| metric 公制 | mAP@.5 | mAP@[.5, .95] |
|---|---|---|
| VGG-16 | 41.5 | 21.2 |
| ResNet-101 | **48.4** | **27.2** |

Table 8: Object detection mAP (%) on the COCO validation set using **baseline** Faster R-CNN. See also Table 9 for better results.

表 8： 使用**基线** Faster R-CNN 的 COCO 验证集对象检测 mAP（%）。另见表 9 以获得更好的结果。

Our method has good generalization performance on other recognition tasks. Table 8 and 8 show the object detection baseline results on PASCAL VOC 2007 and 2012 [5] and COCO [26]. We adopt *Faster R-CNN* [32] as the detection method. Here we are interested in the improvements of replacing VGG-16 [41] with ResNet-101. The detection implementation (see appendix) of using both models is the same, so the gains can only be attributed to better networks. Most remarkably, on the challenging COCO dataset we obtain a 6.0% increase in COCO’s standard metric (mAP@[.5, .95]), which is a 28% relative improvement. This gain is solely due to the learned representations.

我们的方法在其他识别任务中具有良好的泛化性能。表 8 和表 8 展示了 PASCAL VOC 2007 年和 2012 年[5] 以及 COCO [26] 的物体检测基线结果。我们采用 *Faster R-CNN*[32] 作为检测方法。这里我们关注的是用 ResNet-101 替代 VGG-16 [41] 的改进。使用两种模型的检测实现（见附录）相同，因此增益只能归因于更好的网络。最显著的是，在具有挑战性的 COCO 数据集上，COCO 标准指标（mAP@[.5， .95]）提升了 6.0%，相对提升 28%。这种增益完全来自于学习到的表示。

Based on deep residual nets, we won the 1st places in several tracks in ILSVRC & COCO 2015 competitions: ImageNet detection, ImageNet localization, COCO detection, and COCO segmentation. The details are in the appendix.

基于深度残留网，我们在 ILSVRC 和 COCO 2015 竞赛中多个赛道获得第一名：ImageNet 检测、ImageNet 定位、COCO 检测和 COCO 分割。详细信息见附录。

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## Appendix A Object Detection Baselines
## 附录 A 物体探测基线

In this section we introduce our detection method based on the baseline Faster R-CNN [32] system. The models are initialized by the ImageNet classification models, and then fine-tuned on the object detection data. We have experimented with ResNet-50/101 at the time of the ILSVRC & COCO 2015 detection competitions.

本节介绍基于基线 Faster R-CNN [32] 系统的检测方法。模型由 ImageNet 分类模型初始化，然后根据物体检测数据进行微调。我们在 ILSVRC 和 COCO 2015 检测竞赛期间进行了 ResNet-50/101 的试验。

Unlike VGG-16 used in [32], our ResNet has no hidden fc layers. We adopt the idea of “Networks on Conv feature maps” (NoC) [33] to address this issue. We compute the full-image shared conv feature maps using those layers whose strides on the image are no greater than 16 pixels (*i.e.*, conv1, conv2_x, conv3_x, and conv4_x, totally 91 conv layers in ResNet-101; Table 1). We consider these layers as analogous to the 13 conv layers in VGG-16, and by doing so, both ResNet and VGG-16 have conv feature maps of the same total stride (16 pixels). These layers are shared by a region proposal network (RPN, generating 300 proposals) [32] and a Fast R-CNN detection network [7]. RoI pooling [7] is performed before conv5_1. On this RoI-pooled feature, all layers of conv5_x and up are adopted for each region, playing the roles of VGG-16’s fc layers. The final classification layer is replaced by two sibling layers (classification and box regression [7]).

与 [32] 中使用的 VGG-16 不同，我们的 ResNet 没有隐藏的 fc 层。我们采用“网络在卷积特征图”（NoC）[33] 的理念来解决这个问题。我们使用图像步幅不超过 16 像素的图层（ *即* conv1、conv2_x、conv3_x 和 conv4_x，ResNet-101 共 91 个卷积层;表 1）计算全图像共享卷积特征映射。我们认为这些图层类似于 VGG-16 中的 13 个卷积层，因此 ResNet 和 VGG-16 的卷积特征图具有相同的总步幅（16 像素）。这些层由区域提议网络（RPN，产生 300 个提案）[32] 和快速 R-CNN 检测网络 [7] 共享。投资回报率池 [7] 在 conv5_1 之前完成。在这个 RoI 池特征中，每个区域都采用了所有 conv5_x 及以上层，扮演 VGG-16 的 FC 层的角色。最终的分类层被两个兄弟层（分类和盒回归 [7]）所取代。

For the usage of BN layers, after pre-training, we compute the BN statistics (means and variances) for each layer on the ImageNet training set. Then the BN layers are fixed during fine-tuning for object detection. As such, the BN layers become linear activations with constant offsets and scales, and BN statistics are not updated by fine-tuning. We fix the BN layers mainly for reducing memory consumption in Faster R-CNN training.

对于 BN 层的使用，在预训练后，我们计算 ImageNet 训练集中每个图层的 BN 统计量（均值和方差）。然后在微调物体检测时，BN 层被固定。因此，BN 层成为具有恒定偏移和尺度的线性激活，BN 统计不会通过微调更新。我们主要修复 BN 层以减少快速 R-CNN 训练中的内存消耗。

**PASCAL VOC 帕斯卡尔 VOC**

Following [7, 32], for the PASCAL VOC 2007 *test* set, we use the 5k *trainval* images in VOC 2007 and 16k *trainval* images in VOC 2012 for training (“07+12”). For the PASCAL VOC 2012 *test* set, we use the 10k *trainval*+*test* images in VOC 2007 and 16k *trainval* images in VOC 2012 for training (“07++12”). The hyper-parameters for training Faster R-CNN are the same as in [32]. Table 8 shows the results. ResNet-101 improves the mAP by >3% over VGG-16. This gain is solely because of the improved features learned by ResNet.

继 [7， 32] 之后，对于 PASCAL VOC 2007 *测试*集，我们使用 VOC 2007 中的 5k *列车阀*图像和 VOC 2012 中的 16k *列车阀*图像（“07+12”）进行训练。对于 PASCAL VOC 2012 *测试*集，我们在 VOC 2007 中使用 10k *列车阀* + *测试*图像，在 VOC 2012 中使用 16k *列车阀*图像进行训练（“07++12”）。训练更快 R-CNN 的超参数与 [32] 中相同。表 8 显示了结果。ResNet-101 比 VGG-16 提升了 > 3% 的 mAP。这一提升完全是因为 ResNet 学习到的改进功能。

**MS COCO 可可女士**

The MS COCO dataset [26] involves 80 object categories. We evaluate the PASCAL VOC metric (mAP @ IoU = 0.5) and the standard COCO metric (mAP @ IoU = .5:.05:.95). We use the 80k images on the train set for training and the 40k images on the val set for evaluation. Our detection system for COCO is similar to that for PASCAL VOC. We train the COCO models with an 8-GPU implementation, and thus the RPN step has a mini-batch size of 8 images (*i.e.*, 1 per GPU) and the Fast R-CNN step has a mini-batch size of 16 images. The RPN step and Fast R-CNN step are both trained for 240k iterations with a learning rate of 0.001 and then for 80k iterations with 0.0001.

MS COCO 数据集[26] 包含 80 个对象类别。我们评估 PASCAL 度量（mAP @ IoU = 0.5）和标准 COCO 指标（mAP @ IoU = 0.5： .05： .95）。我们用火车组上的 80k 图像进行训练，用 val 组的 40k 图像进行评估。我们的 COCO 检测系统与 PASCAL 挥发性有机化合物类似。我们用 8 GPU 实现训练 COCO 模型，因此 RPN 步骤的迷你批次大小为 8 张图像（ *即*每 GPU1 张），快速 R-CNN 步骤的迷你批次大小为 16 张图像。RPN 步和快速 R-CNN 步均为 240k 迭代，学习率为 0.001，随后 80k 迭代，学习率为 0.0001。

Table 8 shows the results on the MS COCO validation set. ResNet-101 has a 6% increase of mAP@[.5, .95] over VGG-16, which is a 28% relative improvement, solely contributed by the features learned by the better network. Remarkably, the mAP@[.5, .95]’s absolute increase (6.0%) is nearly as big as mAP@.5’s (6.9%). This suggests that a deeper network can improve both recognition and localization.

表 8 显示了 MS COCO 验证集的结果。ResNet-101 比 VGG-16 提升了 6% 的 mAP@[.5， .95]，相对提升 28%，完全得益于更优网络学到的特性。值得注意的是，mAP@[.5， .95]的绝对增长（6.0%）几乎与 mAP@.5（6.9%）相当。这表明更深的网络可以同时提升识别和定位。

## Appendix B Object Detection Improvements
## 附录 B 物体检测改进

For completeness, we report the improvements made for the competitions. These improvements are based on deep features and thus should benefit from residual learning.

为了完整性，我们报告了比赛的改进情况。这些改进基于深度特征，因此应从残留学习中受益。

| training data 训练数据 | COCO train COCO 列车 | | COCO trainval COCO 列车阀 | |
|---|---|---|---|---|
| test data 测试数据 | COCO val 可可谷 | | COCO test-dev COCO 测试开发 | |
| mAP | @.5 | @[.5, .95] | @.5 | @[.5, .95] |
| baseline Faster R-CNN (VGG-16)<br>基线 Faster R-CNN (VGG-16) | 41.5 | 21.2 | | |
| baseline Faster R-CNN (ResNet-101)<br>基线 Faster R-CNN (ResNet-101) | 48.4 | 27.2 | | |
| +box refinement +盒细化 | 49.9 | 29.9 | | |
| +context +背景 | 51.1 | 30.0 | 53.3 | 32.2 |
| +multi-scale testing +多尺度测试 | 53.8 | 32.5 | 55.7 | 34.9 |
| ensemble 一起 | | | **59.0** | **37.4** |

Table 9: Object detection improvements on MS COCO using Faster R-CNN and ResNet-101.

表 9： 利用更快的 R-CNN 和 ResNet-101 对 MS COCO 进行的对象检测改进。

| system 系统 | net 净值 | data 数据 | mAP | aero 阿罗 | bike 自行车 | bird 鸟 | boat 船 | bottle 瓶子 | bus 巴士 | car 汽车 | cat 猫 | chair 椅子 | cow 牛 | table 表格 | dog 狗 | horse 马 | mbike 摩托 | person 人物 | plant 工厂 | sheep 羊 | sofa 沙发 | train 列车 | tv 电视 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| baseline 基线 | VGG-16 | 07+12 | 73.2 | 76.5 | 79.0 | 70.9 | 65.5 | 52.1 | 83.1 | 84.7 | 86.4 | 52.0 | 81.9 | 65.7 | 84.8 | 84.6 | 77.5 | 76.7 | 38.8 | 73.6 | 73.9 | 83.0 | 72.6 |
| baseline 基线 | ResNet-101 | 07+12 | 76.4 | 79.8 | 80.7 | 76.2 | 68.3 | 55.9 | 85.1 | 85.3 | **89.8** | 56.7 | 87.8 | 69.4 | 88.3 | 88.9 | 80.9 | 78.4 | 41.7 | 78.6 | 79.8 | 85.3 | 72.0 |
| baseline+++ 基线+++ | ResNet-101 | COCO+07+12 | **85.6** | **90.0** | **89.6** | **87.8** | **80.8** | **76.1** | **89.9** | **89.9** | 89.6 | **75.5** | **90.0** | **80.7** | **89.6** | **90.3** | **89.1** | **88.7** | **65.4** | **88.1** | **85.6** | **89.0** | **86.8** |

Table 10: Detection results on the PASCAL VOC 2007 test set. The baseline is the Faster R-CNN system. The system “baseline+++” include box refinement, context, and multi-scale testing in Table 9.

表 10： PASCAL VOC 2007 测试集的检测结果。基线是 Faster R-CNN 系统。系统“baseline+++”包括表 9 中的盒细化、上下文和多尺度测试。

| system 系统 | net 净值 | data 数据 | mAP | aero 阿罗 | bike 自行车 | bird 鸟 | boat 船 | bottle 瓶子 | bus 巴士 | car 汽车 | cat 猫 | chair 椅子 | cow 牛 | table 表格 | dog 狗 | horse 马 | mbike 摩托 | person 人物 | plant 工厂 | sheep 羊 | sofa 沙发 | train 列车 | tv 电视 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| baseline 基线 | VGG-16 | 07++12 | 70.4 | 84.9 | 79.8 | 74.3 | 53.9 | 49.8 | 77.5 | 75.9 | 88.5 | 45.6 | 77.1 | 55.3 | 86.9 | 81.7 | 80.9 | 79.6 | 40.1 | 72.6 | 60.9 | 81.2 | 61.5 |
| baseline 基线 | ResNet-101 | 07++12 | 73.8 | 86.5 | 81.6 | 77.2 | 58.0 | 51.0 | 78.6 | 76.6 | 93.2 | 48.6 | 80.4 | 59.0 | 92.1 | 85.3 | 84.8 | 80.7 | 48.1 | 77.3 | 66.5 | 84.7 | 65.6 |
| baseline+++ 基线+++ | ResNet-101 | COCO+07++12 | **83.8** | **92.1** | **88.4** | **84.8** | **75.9** | **71.4** | **86.3** | **87.8** | **94.2** | **66.8** | **89.4** | **69.2** | **93.9** | **91.9** | **90.9** | **89.6** | **67.9** | **88.2** | **76.8** | **90.3** | **80.0** |

Table 11: Detection results on the PASCAL VOC 2012 test set (http://host.robots.ox.ac.uk:8080/leaderboard/displaylb.php?challengeid=11&compid=4). The baseline is the Faster R-CNN system. The system “baseline+++” include box refinement, context, and multi-scale testing in Table 9.

表 11： PASCAL VOC 2012 测试集（http://host.robots.ox.ac.uk:8080/leaderboard/displaylb.php?challengeid=11&compid=4）检测结果。基线是 Faster R-CNN 系统。系统“baseline+++”包括表 9 中的盒细化、上下文和多尺度测试。

**MS COCO 可可女士**

*Box refinement.* Our box refinement partially follows the iterative localization in [6]. In Faster R-CNN, the final output is a regressed box that is different from its proposal box. So for inference, we pool a new feature from the regressed box and obtain a new classification score and a new regressed box. We combine these 300 new predictions with the original 300 predictions. Non-maximum suppression (NMS) is applied on the union set of predicted boxes using an IoU threshold of 0.3 [8], followed by box voting [6]. Box refinement improves mAP by about 2 points (Table 9).

*盒子精炼。* 我们的盒细化部分遵循 [6] 中的迭代定位。在 Faster R-CNN 中，最终输出是一个与其提案框不同的回归框。因此，为了推理，我们从回归框中汇集一个新特征，获得新的分类分数和新的回归框。我们将这 300 个新预测与原 300 个预测结合。非最大抑制（NMS）在预测框的并集上使用 IoU 阈值 0.3 [8]，随后进行盒投票 [6]。盒体精炼使 mAP 提升约 2 个百分点（见表 9）。

*Global context.* We combine global context in the Fast R-CNN step. Given the full-image conv feature map, we pool a feature by global Spatial Pyramid Pooling [12] (with a “single-level” pyramid) which can be implemented as “RoI” pooling using the entire image’s bounding box as the RoI. This pooled feature is fed into the post-RoI layers to obtain a global context feature. This global feature is concatenated with the original per-region feature, followed by the sibling classification and box regression layers. This new structure is trained end-to-end. Global context improves mAP@.5 by about 1 point (Table 9).

*全球背景。* 我们在快速 R-CNN 步骤中结合了全球背景。给定全图像卷积特征图，我们通过全局空间金字塔池化[12]（采用“单层”金字塔）来汇集一个特征，可以用整个图像的边界框作为 RoI 实现“投资回报率”池。该汇集特征被输入到后投资回报率层，以获得全局上下文特征。该全局特征与原始的按区域特征串联，随后是兄弟分类层和盒回归层。这种新结构经过端到端训练。全球背景使 mAP@.5 提升约 1 分（见表 9）。

*Multi-scale testing.* In the above, all results are obtained by single-scale training/testing as in [32], where the image’s shorter side is $s=600$ pixels. Multi-scale training/testing has been developed in [12, 7] by selecting a scale from a feature pyramid, and in [33] by using maxout layers. In our current implementation, we have performed multi-scale *testing* following [33]; we have not performed multi-scale training because of limited time. In addition, we have performed multi-scale testing only for the Fast R-CNN step (but not yet for the RPN step). With a trained model, we compute conv feature maps on an image pyramid, where the image’s shorter sides are $s\in\{200,400,600,800,1000\}$. We select two adjacent scales from the pyramid following [33]. RoI pooling and subsequent layers are performed on the feature maps of these two scales [33], which are merged by maxout as in [33]. Multi-scale testing improves the mAP by over 2 points (Table 9).

*多尺度测试。* 在上述中，所有结果均通过单尺度训练/测试获得，如 [32] 所示，图像短边为 $s=600$ 像素。多尺度训练/测试在 [12， 7] 中通过从特征金字塔中选择尺度开发，在 [33] 中使用最大化层。在我们当前的实现中，我们在 [33] 之后进行了多尺度*测试*;由于时间有限，我们没有进行多尺度训练。此外，我们只针对快速 R-CNN 步骤进行了多尺度测试（但尚未针对 RPN 步骤）。使用训练好的模型，我们在图像金字塔上计算卷积特征映射，图像的短边为 $s\in\{200,400,600,800,1000\}$。我们从金字塔中选择两个相邻的尺度，跟随[33]。RoI 池和后续层在这两个尺度的特征图上执行 [33]，并通过 maxout 合并，如 [33]。多尺度测试使 mAP 提升了 2 个百分点以上（见表 9）。

*Using validation data.* Next we use the 80k+40k trainval set for training and the 20k test-dev set for evaluation. The test-dev set has no publicly available ground truth and the result is reported by the evaluation server. Under this setting, the results are an mAP@.5 of 55.7% and an mAP@[.5, .95] of 34.9% (Table 9). This is our single-model result.

*使用验证数据。* 接下来我们使用 80k+40k 训练集进行训练，20k 测试开发集用于评估。测试开发集没有公开的真实数据，结果由评估服务器报告。在此设置下，结果为 mAP@.5 为 55.7%，mAP@[.5， .95] 为 34.9%（见表 9）。这是我们的单模型结果。

*Ensemble.* In Faster R-CNN, the system is designed to learn region proposals and also object classifiers, so an ensemble can be used to boost both tasks. We use an ensemble for proposing regions, and the union set of proposals are processed by an ensemble of per-region classifiers. Table 9 shows our result based on an ensemble of 3 networks. The mAP is 59.0% and 37.4% on the test-dev set. *This result won the 1st place in the detection task in COCO 2015.*

*合奏。* 在 Faster R-CNN 中，系统设计用于学习区域提案和对象分类器，因此可以用集合来提升这两项任务。我们使用一个集合来提出区域，提案的并集由每个区域分类器组成的集合处理。表 9 展示了基于 3 个网络组合的结果。测试开发集中的 mAP 分别为 59.0%和 37.4%。 *该结果在 COCO 2015 年检测任务中获得第一名。*

**PASCAL VOC 帕斯卡尔 VOC**

We revisit the PASCAL VOC dataset based on the above model. With the single model on the COCO dataset (55.7% mAP@.5 in Table 9), we fine-tune this model on the PASCAL VOC sets. The improvements of box refinement, context, and multi-scale testing are also adopted. By doing so we achieve 85.6% mAP on PASCAL VOC 2007 (Table 10) and 83.8% on PASCAL VOC 2012 (Table 11)[6]
我们基于上述模型重新审视 PASCAL VOC 数据集。利用 COCO 数据集上的单一模型（表 9 中 55.7% mAP@.5），我们在 PASCAL VOC 集合上对该模型进行了微调。同时采用了盒子细化、上下文和多尺度测试的改进。通过这样做，我们在 PASCAL VOC 2007（表 10）上实现了 85.6%的 mAP，在 PASCAL VOC 2012（表 11）上实现了 83.8%。6. The result on PASCAL VOC 2012 is 10 points higher than the previous state-of-the-art result [6].
.PASCAL VOC 2012 的结果比之前的先进结果高出 10 个百分点 [6]。

[6] http://host.robots.ox.ac.uk:8080/anonymous/3OJ4OJ.html, submitted on 2015-11-26.
http://host.robots.ox.ac.uk:8080/anonymous/3OJ4OJ.html，提交于 2015-11-26。

**ImageNet Detection ImageNet 检测**

| | val2 | test 测试 |
|---|---|---|
| GoogLeNet [44] (ILSVRC'14)<br>GoogLeNet [44] (ILSVRC'14) | - | 43.9 |
| our single model (ILSVRC'15)<br>我们的单一模型 (ILSVRC'15) | 60.5 | 58.8 |
| our ensemble (ILSVRC'15) 我们的合奏组 (ILSVRC'15) | **63.6** | **62.1** |

Table 12: Our results (mAP, %) on the ImageNet detection dataset. Our detection system is Faster R-CNN [32] with the improvements in Table 9, using ResNet-101.

表 12： 我们在 ImageNet 检测数据集上的结果（mAP，%）。我们的检测系统是 Faster R-CNN [32]，采用表 9 改进，采用 ResNet-101 技术。

The ImageNet Detection (DET) task involves 200 object categories. The accuracy is evaluated by mAP@.5. Our object detection algorithm for ImageNet DET is the same as that for MS COCO in Table 9. The networks are pre-trained on the 1000-class ImageNet classification set, and are fine-tuned on the DET data. We split the validation set into two parts (val1/val2) following [8]. We fine-tune the detection models using the DET training set and the val1 set. The val2 set is used for validation. We do not use other ILSVRC 2015 data. Our single model with ResNet-101 has 58.8% mAP and our ensemble of 3 models has 62.1% mAP on the DET test set (Table 12). *This result won the 1st place in the ImageNet detection task in ILSVRC 2015*, surpassing the second place by **8.5 points** (absolute).

ImageNet 检测（DET）任务涉及 200 个对象类别。准确度以 mAP@.5 进行评估。我们针对 ImageNet DET 的对象检测算法与表 9 中 MS COCO 的算法相同。这些网络在 1000 类 ImageNet 分类集上预训练，并根据 DET 数据进行微调。我们将验证集分为两部分（val1/val2），遵循 [8]。 我们使用 DET 训练集和 val1 集对检测模型进行微调。val2 集用于验证。我们不使用其他 ILSVRC 2015 数据。我们单一模型 ResNet-101 的 mAP 为 58.8%，我们由 3 个模型组成的 DET 测试集为 62.1% mAP（见表 12）。 *该结果在 ILSVRC 2015 中获得 ImageNet 检测任务第一名*，比第二名高出 **8.5 个百分点**（绝对分数）。

## Appendix C ImageNet Localization
## 附录 C ImageNet 本地化

| LOC method 方法 | LOC network 网络 | testing 测试 | LOC error LOC 错误 on GT CLS 在 GT CLS 上 | classification 分类 network 网络 | top-5 LOC error 前五名 LOC 错误 on predicted CLS 关于预测的 CLS |
|---|---|---|---|---|---|
| VGG's [41]<br>VGG 的 [41] | VGG-16 | 1-crop 1-作物 | 33.1 [41] | | |
| RPN | ResNet-101 | 1-crop 1-作物 | 13.3 | | |
| RPN | ResNet-101 | dense 密集 | 11.7 | | |
| RPN | ResNet-101 | dense 密集 | | ResNet-101 | 14.4 |
| RPN+RCNN | ResNet-101 | dense 密集 | | ResNet-101 | **10.6** |
| RPN+RCNN | ensemble 一起 | dense 密集 | | ensemble 一起 | **8.9** |

Table 13: Localization error (%) on the ImageNet validation. In the column of “LOC error on GT class” ([41]), the ground truth class is used. In the “testing” column, “1-crop” denotes testing on a center crop of 224×224 pixels, “dense” denotes dense (fully convolutional) and multi-scale testing.

表 13： ImageNet 验证中的定位误差（%）。在“GT 类别的 LOC 误差”一栏（[41]）中，使用了地面真实类别。在“测试”栏中，“1-裁剪”表示在中心裁剪为 224×\times224 像素的测试，“dense”表示密集（完全卷积）和多尺度测试。

The ImageNet Localization (LOC) task [36] requires to classify and localize the objects. Following [40, 41], we assume that the image-level classifiers are first adopted for predicting the class labels of an image, and the localization algorithm only accounts for predicting bounding boxes based on the predicted classes. We adopt the “per-class regression” (PCR) strategy [40, 41], learning a bounding box regressor for each class. We pre-train the networks for ImageNet classification and then fine-tune them for localization. We train networks on the provided 1000-class ImageNet training set.

ImageNet 定位（LOC）任务 [36] 需要对对象进行分类和定位。根据 [40， 41]，我们假设图像级分类器首先被用于预测图像的类标签，而局部化算法只考虑基于预测类别的边界框预测。我们采用“逐类回归”（PCR）策略 [40， 41]，为每个类学习一个边界框回归器。我们先为 ImageNet 分类预训练网络，然后进行局部化微调。我们在提供的 1000 类 ImageNet 训练集上训练网络。

Our localization algorithm is based on the RPN framework of [32] with a few modifications. Unlike the way in [32] that is category-agnostic, our RPN for localization is designed in a *per-class* form. This RPN ends with two sibling 1×1 convolutional layers for binary classification (*cls*) and box regression (*reg*), as in [32]. The *cls* and *reg* layers are both in a *per-class* from, in contrast to [32]. Specifically, the *cls* layer has a 1000-d output, and each dimension is *binary logistic regression* for predicting being or not being an object class; the *reg* layer has a 1000×4-d output consisting of box regressors for 1000 classes. As in [32], our bounding box regression is with reference to multiple translation-invariant “anchor” boxes at each position.

我们的定位算法基于 [32] 的 RPN 框架，并做了一些修改。与 [32] 中类别无关的方式不同，我们的局部化 RPN 是以*逐类*形式设计的。该 RPN 以两个兄弟层 1 × 1 卷积层结束，分别用于二元分类（*cls*）和盒回归（*reg*），如 [32 ]。*CLS* 层和 *reg* 层都属于*每个类别*，与 [32] 形成对比。具体来说，*cls* 层有 1000 维输出，每个维度都是二元*逻辑回归*，用于预测是否是对象类;Reg Layer 有 1000 × 个 4D 输出，包含 1000 类的盒式回归器。如同[32]，我们的包围盒回归是参考每个位置多个平移不变“锚”盒。

As in our ImageNet classification training (Sec. 3.4), we randomly sample 224×224 crops for data augmentation. We use a mini-batch size of 256 images for fine-tuning. To avoid negative samples being dominate, 8 anchors are randomly sampled for each image, where the sampled positive and negative anchors have a ratio of 1:1 [32]. For testing, the network is applied on the image fully-convolutionally.

与我们的 ImageNet 分类训练（第 3.4 节）一样，我们随机抽样 224 × 224 种作物进行数据补充。我们使用 256 张小批量图像进行微调。为避免负样本占主导，每张图像随机抽样 8 个锚点，其中正负锚点的比例为 1：1[32 ]。测试时，网络以全卷积方式应用于图像。

| method 方法 | top-5 localization err 前五名本地化 错误<br>val 瓦尔 | test 测试 |
|---|---|---|
| OverFeat [40] (ILSVRC'13)<br>超功 [40] (ILSVRC'13) | 30.0 | 29.9 |
| GoogLeNet [44] (ILSVRC'14)<br>GoogLeNet [44] (ILSVRC'14) | - | 26.7 |
| VGG [41] (ILSVRC'14)<br>VGG [41] (ILSVRC'14) | 26.9 | 25.3 |
| ours (ILSVRC'15) 我 (ILSVRC'15) | **8.9** | **9.0** |

Table 14: Comparisons of localization error (%) on the ImageNet dataset with state-of-the-art methods.

表 14： ImageNet 数据集定位误差（%）与先进方法的比较。

Table 13 compares the localization results. Following [41], we first perform “oracle” testing using the ground truth class as the classification prediction. VGG’s paper [41] reports a center-crop error of 33.1% (Table 13) using ground truth classes. Under the same setting, our RPN method using ResNet-101 net significantly reduces the center-crop error to 13.3%. This comparison demonstrates the excellent performance of our framework. With dense (fully convolutional) and multi-scale testing, our ResNet-101 has an error of 11.7% using ground truth classes. Using ResNet-101 for predicting classes (4.6% top-5 classification error, Table 5), the top-5 localization error is 14.4%.

表 13 比较了本地化结果。按照[41]，我们首先使用地面真实类作为分类预测进行“预言”测试。VGG 的论文 [41] 报告，使用地面真实类别的中心作物误差为 33.1%（见表 13）。在同一环境下，我们使用 ResNet-101 网的 RPN 方法显著将中心作物误差降至 13.3%。此对比展示了我们框架的卓越性能。通过密集（全卷积）和多尺度测试，我们的 ResNet-101 使用地面真实类时误差为 11.7%。使用 ResNet-101 预测类别（前五分类误差 4.6%，见表 5），前五名定位误差为 14.4%。

The above results are only based on the *proposal network* (RPN) in Faster R-CNN [32]. One may use the *detection network* (Fast R-CNN [7]) in Faster R-CNN to improve the results. But we notice that on this dataset, one image usually contains a single dominate object, and the proposal regions highly overlap with each other and thus have very similar RoI-pooled features. As a result, the image-centric training of Fast R-CNN [7] generates samples of small variations, which may not be desired for stochastic training. Motivated by this, in our current experiment we use the original R-CNN [8] that is RoI-centric, in place of Fast R-CNN.

上述结果仅基于 Faster R-CNN 中的*提案网络*（RPN）[32]。可以在《更快的 R-CNN》中使用*检测网络*（Fast R-CNN [7]）来改进结果。但我们注意到在这个数据集上，一张图像通常包含一个主导对象，提案区域高度重叠，因此具有非常相似的投资回报率池特征。因此，快速 R-CNN 的图像中心训练[7] 会产生微小变化的样本，这对于随机训练来说可能并不理想。基于此，在我们当前的实验中，我们使用了以 RoI 为中心的原始 R-CNN [8]，取代了快速 R-CNN。

Our R-CNN implementation is as follows. We apply the per-class RPN trained as above on the training images to predict bounding boxes for the ground truth class. These predicted boxes play a role of class-dependent proposals. For each training image, the highest scored 200 proposals are extracted as training samples to train an R-CNN classifier. The image region is cropped from a proposal, warped to 224×224 pixels, and fed into the classification network as in R-CNN [8]. The outputs of this network consist of two sibling fc layers for *cls* and *reg*, also in a per-class form. This R-CNN network is fine-tuned on the training set using a mini-batch size of 256 in the RoI-centric fashion. For testing, the RPN generates the highest scored 200 proposals for each predicted class, and the R-CNN network is used to update these proposals’ scores and box positions.

我们的 R-CNN 实现如下。我们应用上述训练的每类 RPN 在训练图像上，预测真实类的边界框。这些预测框起到了类依赖提案的作用。对于每张训练图像，最高分的 200 个提案被提取作为训练样本，用于训练 R-CNN 分类器。图像区域从提案中裁剪，扭曲到 × 224,224 像素，并输入到网络，如 R-CNN[8]。 该网络的输出由两个兄弟 fc 层组成，分别用于 *cls* 和 *reg*，同样以每类形式存在。该 R-CNN 网络在训练集上采用以投资回报率为中心的 256 个小批量进行微调。在测试中，RPN 为每个预测类别生成最高分的 200 个提案，并利用 R-CNN 网络更新这些提案的得分和框位置。

This method reduces the top-5 localization error to 10.6% (Table 13). This is our single-model result on the validation set. Using an ensemble of networks for both classification and localization, we achieve a top-5 localization error of 9.0% on the test set. This number significantly outperforms the ILSVRC 14 results (Table 14), showing a 64% relative reduction of error. *This result won the 1st place in the ImageNet localization task in ILSVRC 2015.*

该方法将前五名定位误差降低到 10.6%（见表 13）。这是我们在验证集上的单模型结果。通过使用一组网络进行分类和定位，我们在测试集上实现了 9.0% 的前五名定位误差。这一数字显著优于 ILSVRC 14 的结果（见表 14），显示误差相对减少了 64%。 *该结果在 2015 年 ILSVRC 的 ImageNet 本地化任务中获得第一名。*

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